

Investigation of Delayed Neutron Precursor Data Uncertainties for Microscopic and Summation Calculations

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Abstract

This paper presents an analysis on the sensitivity of [delayed neutron precursor \(DNP\)](#) group parameters, group half-lives τ_k and yields $\nu_{d,k}$, and summed delayed neutron yields to uncertainties in their nuclear data for various datasets, including the [International Atomic Energy Agency \(IAEA\)](#) beta-delayed neutron emission database, ENDF/B-VII.1, ENDF/B-VIII.0, JEFF-3.1.1, and JENDL-5. These uncertainty-weighted sensitivities provide a detailed view of which [DNPs](#) need more accurate data such that analyses using individual [DNPs](#), such as microscopic group parameter formulation and summation calculations, may be more accurate. Improving the accuracy of these analyses is of particular importance for kinetics models, which rely on accurate [DNP](#) group parameters. The microscopic approach for group parameter calculation is useful for providing insight into how each [DNP](#) contributes to delayed neutron emission, but it requires accurate nuclear data. The analyses conducted in this work reveal several [DNPs](#) that correlate strongly with the [DNP](#) group parameters, indicating a need for more accurate data for those [DNPs](#). There are also [DNPs](#) which have large uncertainties and vary significantly between datasets which also warrant additional attention. This paper informs which [DNPs](#) are in need of updates to their data for various nuclear datasets.

Keywords — Delayed Neutron Precursors, Group Parameters, Nuclear Data, Nuclear Data Uncertainties, ENDF, JEFF

I. INTRODUCTION

Modeling and simulation of nuclear reactors enables more accurate predictions about the reactor behavior during transient events, such as removal of a control rod. To accurately model these dynamics, several different effects must be included: the change in the neutron density, the various feedback effects that influence the reactivity, the emission of prompt neutrons, and the emission of delayed neutrons. These effects are well understood and have been simplified using various approximations. For example, the delayed neutrons are treated as having six to eight sources, or **DNP** groups, even though in reality there are hundreds of precursor nuclides that decay and emit these delayed neutrons [1].

There are several different methods for calculating these **DNP** groups, and in this work the microscopic approach is of particular interest [2, 3, 4, 5, 6, 7, 8, 9]. This approach uses experimental data for individual **DNPs** combined together computationally. One strength of this approach is that it provides insight into how each individual **DNP** contributes to the **DNP** group parameters. However, this strength is also a weakness, as it relies on the nuclear data for every **DNP** individually, which means a few large uncertainties in the various **DNPs** can lead to uncertain results for the **DNP** group parameters. There is also a related, and similar approach, called the summation approach. This approach uses the same data, and is used for calculation of summed parameters such as the total delayed neutron yield ν_d and the average half-life $\bar{\tau}$.

There have been investigations in the literature that use the microscopic and summation approaches to evaluate sensitivities and compare different datasets [10, 9]. Although these analyses are informative and show many of the differences between datasets, they do not discuss the importance of **DNPs** in regards to their effect on the calculated **DNP** group parameters but instead based on their contribution to the total delayed neutron yield. Generally the larger the delayed neutron yield of a **DNP**, the more important it is, but these analyses do not investigate how the uncertainty in each **DNP** propagates through to calculated **DNP** group parameters.

This work seeks to expand upon these previous analyses by using the microscopic approach in combination with Monte Carlo stochastic sampling of nuclear data uncertainties to evaluate uncertainty-weighted sensitivities of individual **DNP**'s to **DNP** group half-lives and delayed neutron yields. These results indicate which **DNPs** in different datasets have large impacts on the **DNP** group parameters, as well as which **DNP** have both large impacts and large uncertainties relative

to those impacts. By providing the data of importance for these datasets, future analyses can target experiments to produce those specific data to further improve DNP group parameters and kinetics models.

II. METHODS

This section details the three stages to the analyses presented in this paper: the data treated as an input, the solver which calculates the data of interest, and the data analysis which details how the solver results are processed. The solver itself can be considered a converter of the DNP nuclear data into six group DNP parameters, which are then passed into data analysis calculations.

II.A. Data

There are two different sets of data used in this work: the nuclear data for each individual DNP and the calculated DNP group parameters that approximate those individual DNPs. The I DNP have the following data:

CFY_i = the cumulative fission yield of DNP i [-]

$P_{n,i}$ = the emission probability of DNP i [-]

τ_i = the half-life of DNP i [s]

$\chi_{d,i}$ = the delayed neutron energy spectrum from DNP i [-].

The cumulative fission yield represents the total production of that nuclide for a single fission, including decays that eventually lead to that nuclide. The emission probability is the probability for a delayed neutron to be emitted from the DNP^a. For this work, the delayed neutron energy spectra are not investigated, as it is considered out of scope.

The DNP group parameters have the following data:

$\nu_{d,k}$ = the delayed neutron yield of group k [-]

τ_k = the half-life of group k [s]

$\chi_{d,k}$ = the delayed neutron energy spectrum from group k [-].

^aThis can alternatively be considered the number of delayed neutrons emitted per radioactive decay.

The delayed neutron yield for each group $\nu_{d,k}$ combines the emission probabilities and cumulative fission yields, representing the number of delayed neutrons emitted from group k from a single fission.

The datasets used in this work are given in Table I. The IAEA beta-delayed neutron emission database does not have cumulative fission yield data combined with it, though analyses using the data use JEFF-3.1.1 cumulative fission yields [11]. For the analyses in this work, the cumulative fissions yields from all of the datasets will be used and compared against to offer a comprehensive analysis.

TABLE I
Various data sources as well as the data which they contain.

Data name	Year	Source	Available Data		
			τ_i	$P_{n,i}$	CFY_i
IAEA database	2018	[11]	Yes	Yes	No
Joint Evaluated Fission and Fusion (JEFF) 3.1.1	2017	[12]	Yes	Yes	Yes
Evaluated Nuclear Data File (ENDF)/B-VII.1	2011	[13]	Yes	Yes	Yes
ENDF/B-VIII.0	2018	[14]	Yes	Yes	Yes
Japanese Evaluated Nuclear Data Library (JENDL)-5	2021	[15]	Yes	Yes	Yes

The data of interest in these datasets are for the individual DNP nuclear data. This data is passed into the solver, which then converts these data into DNP group parameters for analysis and post-processing.

II.B. Solver

For the microscopic and summation methods, there are two distinct solution methods. The summation method is the simpler of the two, and simply involves summing the nuclear data to calculate the total delayed neutron yield and average half-life. The microscopic method is more in-depth, as it is used to calculate the DNP group parameters, which give time-dependency and can be used to calculate the total delayed neutron yield and average half-life.

Equations (1) and (2) are the equations for the total delayed neutron yield and average

half-life, respectively, calculated using the summation approach as:

$$\nu_d = \sum_{i=1}^I P_{n,i} CFY_i, \quad (1)$$

and

$$\bar{\tau} = \frac{1}{\nu_d} \sum_{i=1}^I CFY_i P_{n,i} \tau_i. \quad (2)$$

The uncertainties for each calculation can be propagated as shown in Equations (3) and (4) as:

$$\Delta \nu_d^2 = \sum_{i=1}^I (\Delta P_{n,i} CFY_i)^2 + (P_{n,i} \Delta CFY_i)^2, \quad (3)$$

and

$$\begin{aligned} \Delta \bar{\tau}^2 = & \frac{1}{\nu_d^2} \sum_{i=1}^I (\Delta CFY_i P_{n,i} \tau_i)^2 + (CFY_i \Delta P_{n,i} \tau_i)^2 \\ & + (CFY_i P_{n,i} \Delta \tau_i)^2 + \frac{1}{\nu_d^2} (CFY_i P_{n,i} \tau_i \Delta \nu_d)^2. \end{aligned} \quad (4)$$

These calculations are automated in the solver using the uncertainties Python package [16]. The standard deviation propagated in the summation calculation are given by the dataset used, and any values without a standard deviation or a standard deviation of 0 use a value of 10^{-12} .

The microscopic approach enables calculation of time-dependent parameters by calculating the delayed neutron count rate from each DNP after irradiation of a fissile nuclide. The solver first calculates the concentration of each DNP, followed by the delayed neutron count rate, and finally uses the Scipy optimize module to calculate the six DNP group parameters that optimally fit that delayed neutron count rate [17]. To calculate sensitivities and propagate uncertainties, a Monte Carlo sampling technique is used [9]. The solution is calculated with tolerances of 10^{-11} for ten randomly sampled initial conditions for the nominal nuclear data, which is then passed as an initial guess for each of the sampled Monte Carlo solves. The concentrations have the uncertainty propagated using the uncertainties package, and then in the count rate calculation the concentration N_i , half-life τ_i , and emission probability $P_{n,i}$ are sampled using a normal distribution with a different random value for each DNP.

The concentration of each DNP is calculated assuming the sample has reached equilibrium^b

^bThe parasitic absorption and transmutation cross-sections of the DNPs are negligible and are not included in

[8]:

$$N_i(t = \infty) = \frac{CFY_i}{\lambda_i}, \quad (5)$$

where

$$\lambda_i = \frac{\ln 2}{\tau_i}$$

= the decay constant for [DNP](#) i [s^{-1}].

The delayed neutron count rate can then be calculated as shown in Equation (6) as:

$$\dot{n}_d(t) = \sum_{i=1}^I P_{n,i} N_i(t = \infty) e^{-\lambda_i t}. \quad (6)$$

The count rate is first calculated using the nominal nuclear data before this calculation is repeated using the independently, normally sampled nuclear data for each [DNP](#).

Once the count rate is calculated, a non-linear least squares calculation can be used to determine the combination of [DNP](#) group parameters most optimally fits the calculated delayed neutron count rate. The coefficient matrix and solution vector for the non-linear least squares solve are based on the [DNP](#) group parameter fit for the delayed neutron count rate, given in Equation (7) as [5]:

$$\dot{n}_d(t) = \sum_{k=1}^K \nu_{d,k} e^{-\lambda_k t}. \quad (7)$$

Once the delayed neutron count rate is acquired, an over-determined non-linear least squares approach is used to calculate the group yields and decays [1, 18, 19]. Equation (8) gives the general form of the least squares equation as:

$$A\vec{x} \approx \vec{b}, \quad (8)$$

this equation.

where

A = the coefficient matrix $[-]$

x = the solution vector $[\# \cdot s^{-1}]$

b = the target vector $[\# \cdot s^{-1}]$.

Equation (9) gives the coefficient matrix as:

$$A = \begin{bmatrix} e^{-\lambda_1 t_0} & e^{-\lambda_2 t_0} & \dots & e^{-\lambda_K t_0} \\ e^{-\lambda_1 t_1} & e^{-\lambda_2 t_1} & \dots & e^{-\lambda_K t_1} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-\lambda_1 t_m} & e^{-\lambda_2 t_m} & \dots & e^{-\lambda_K t_m} \end{bmatrix}. \quad (9)$$

The solution vector contains only the group yields, given in Equation (10) as:

$$\vec{x} = \begin{bmatrix} \bar{\nu}_{d,1} \\ \bar{\nu}_{d,2} \\ \vdots \\ \bar{\nu}_{d,K} \end{bmatrix}. \quad (10)$$

The target vector contains the delayed neutron count rate data for m time-steps, shown in Equation (11) as:

$$\vec{b} = \begin{bmatrix} \dot{n}_d(t_0) \\ \dot{n}_d(t_1) \\ \vdots \\ \dot{n}_d(t_m) \end{bmatrix}. \quad (11)$$

Using least squares with a non-relative residual would lead to early time-step count rates dominating due to their larger magnitudes. This is resolved by minimizing a relative residual, given by Equation (12) as:

$$r = \min_x \left\| \frac{\vec{b} - A\vec{x}}{\vec{b}} \right\|_2, \quad (12)$$

where the Euclidean norm, $\|\cdot\|_2$, is defined in Equation 13 as:

$$\|\vec{x}\|_2 = \sqrt{x_1^2 + \cdots + x_n^2}. \quad (13)$$

The relative residual is used instead of a chi-squared statistic because the Monte Carlo sampling method is already in place to calculate and propagate the uncertainties [20].

The **DNP** group parameters can then be used to evaluate the total delayed neutron yield and average half-life, similar to Equations (1) and (2). This is shown in Equations (14) and (15) as:

$$\nu_d \approx \sum_{k=1}^K \nu_{d,k}, \quad (14)$$

and

$$\bar{\tau} \approx \frac{1}{\nu_d} \sum_{k=1}^K \nu_{d,k} \tau_k. \quad (15)$$

These equations are approximate because the number of groups K used is fewer than the total number of **DNPs** I . To differentiate between the calculation of these parameters using I **DNPs** and K groups, the parameters will be written as $\nu_d(I)$ and $\bar{\tau}(I)$ when calculated using I **DNPs** and $\nu_d(K)$ and $\bar{\tau}(K)$ when calculated using K groups.

II.C. Data Analysis

The **Pearson correlation coefficient (PCC)** is used to analyze the **DNP** group parameters calculated from the solver for various unique combinations of sampled **DNP** nuclear data and is calculated using the SciPy stats module [17]. The **PCC** measures the linear correlation between the **DNP** data and the calculated **DNP** group parameters. A measure of 0 indicates there is no correlation, a measure of +1 indicates perfect positive linear correlation, and a measure of -1 indicates perfect negative linear correlation. So, for example, ^{87}Br can be expected to have a **PCC** near +1 for the impact of its half-life τ_i on the half-life of group one^c τ_1 . This is because ^{87}Br is the longest-lived **DNP**^d, so the longest-lived **DNP** group will be highly dependent on its nuclear data.

^cGroup one is the longest-lived **DNP** group.

^dThere are other **DNPs** with longer half-lives, such as ^{211}Tl , ^{210}Tl , and ^{210}Hg , but they live only tens of seconds longer and generally have negligible concentrations.

The **PCC** was selected to measure variation in the group parameters because the sensitivities did not generally have strong non-linear behavior. The non-linear least squares will result in non-linear effects, but linear correlations dominate in the observed sensitivities. The **PCCs** are calculated for each **DNP** group half-life τ_k and delayed neutron yield $\nu_{d,k}$ based on changes in each **DNPs**' concentration N_i , half-life τ_i , and emission probability $P_{n,i}$. This can be represented as $PCC_{i,v,k}$, where i represents the **DNP** i ; v represents the **DNP** value, such as the half-life, emission probability, or concentration; and k represents the **DNP** group.

To improve this analysis, some of this data was combined to form more concise coefficients. The total **PCC** for a specific **DNP** i is given in Equation (16) is represented as:

$$PCC_i = \sum_v \sum_k |PCC_{i,v,k}|, \quad (16)$$

where

v = the **DNP** value, representing the half-life, emission probability,

or concentration of **DNP** i [-]

$PCC_{i,v,g}$ = the **PCC** for **DNP** i and value v with **DNP** group parameter k [-].

These **PCCs** can be combined with the relative uncertainty associated with each **DNP** to capture the nuclides which both impact the **DNP** group parameters most significantly and have large uncertainties. Specifically, this is calculated in Equation (17) as the relative uncertainties times the magnitude of the **PCC** for a **DNP** i and each **DNP** value v as U_i :

$$U_i = \sum_v \frac{\sigma_{i,v}}{x_{i,v}} \sum_k |PCC_{i,v,k}|, \quad (17)$$

where

U_i = the total uncertainty-weighted sensitivity from the contribution of every DNP i [-]

$\sigma_{i,v}$ = the uncertainty in the DNP value v for DNP i [-]

$x_{i,v}$ = the DNP value v , representing the half-life, emission probability,
or concentration of DNP i [-].

Equation (17) shows that the relative uncertainty for each nuclear data component that is randomly sampled, such as the emission probability, is multiplied by the summed magnitudes of the PCC for each DNP group parameter using that data. These values are then all summed, giving a total measure of the relative uncertainty-weighted PCC for each DNP, which indicates how much attention that specific DNP needs. This metric is larger when:

- changes in the DNP's nuclear data cause large changes in the DNP group parameter(s)
- the relative uncertainty in the DNP's nuclear data is large.

This can also be represented for any individual DNP value by removing the summation over v . This is shown in Equation (18) which gives the uncertainty-weighted PCC for a given DNP i and nuclear data v as:

$$U_{i,v} = \frac{\sigma_{i,v}}{x_{i,v}} \sum_k |PCC_{i,v,k}|. \quad (18)$$

III. RESULTS

This section presents the results for different datasets of cumulative fission yields, emission probabilities, and half-lives. Each sub-section will contain the following analyses for each dataset:

- the total delayed neutron yield calculated from summation of DNPs $\nu_d(I)$ and from the group parameters $\nu_d(K)$
- the average half-life calculated from summation of DNPs $\bar{\tau}(I)$ and from the group parameters $\bar{\tau}(K)$
- the six DNPs with the largest yield contributions

TABLE II
Default parameters for each microscopic calculation.

Parameter	Value	Units
Fissile Nuclide	^{235}U	[-]
Incident Neutron Energy	0.0253	[eV]
Final Decay Time	600	[s]
Decay Time Spacing	Logarithmic	[-]
Decay Time Nodes	800	[-]
Sampled Simulations	5000	[-]

- the tabulated ten largest [PCCs](#) and uncertainty-weighted [PCCs](#)
- a chart of nuclides made using [The Advanced Reactor Modeling Interface \(ARMI\)](#) with summed [PCCs](#) and uncertainty-weighted [PCCs](#) [21]
- a bar chart with the most significant uncertainty-weighted [PCCs](#).

Table II shows the default parameters used for each simulation.

As shown in Table I, the [IAEA](#) database does not include cumulative fission yield data. For that analysis, the JENDL-5 cumulative fission yields are used since they are the most recent data. However, part of the analysis will also investigate the effects of hybrid datasets that vary the emission probability, half-life, and cumulative fission yield data.

This work is also verified with total delayed neutron yields from a related [IAEA Coordinated Research Project \(CRP\)](#) analysis [11]. They present a benchmark problem using JEFF-3.1.1 cumulative fission yields with ENDF/B-VIII.0 emission probabilities and half-lives, which is replicated in this work [22, 14]. The four groups in the study are [The French Alternative Energies and Atomic Energy Commission \(CEA\)](#), [The Centre for Energy, Environmental and Technological Research \(CIEMAT\)](#), the [Japan Atomic Energy Agency \(JAEA\)](#), and the University of Nantes, all of whom reported the same result of 0.01609 ± 0.0008 for the total delayed neutron yield from thermal fission of ^{235}U . This agrees with the same result calculated in this work of 0.01609 ± 0.0008 .

For the first set of results, additional information will be included that is not included in the other sections. Specifically, the delayed neutron count rate will be provided as well as a comparison to the count rate from calculated from other [DNP](#) group parameters in the literature. There will also be an example figure demonstrating how the stochastic sampling affects the [DNP](#) group parameters. Additionally, specific [DNP](#) examples will be provided demonstrating the [PCCs](#).

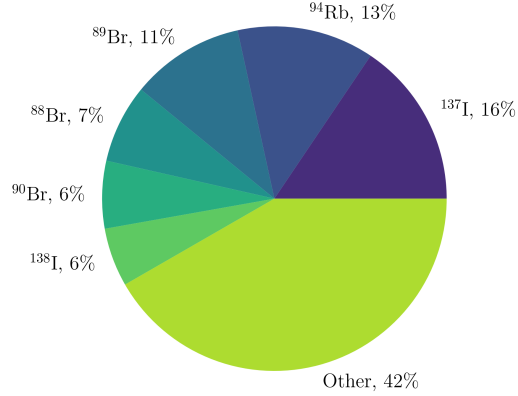


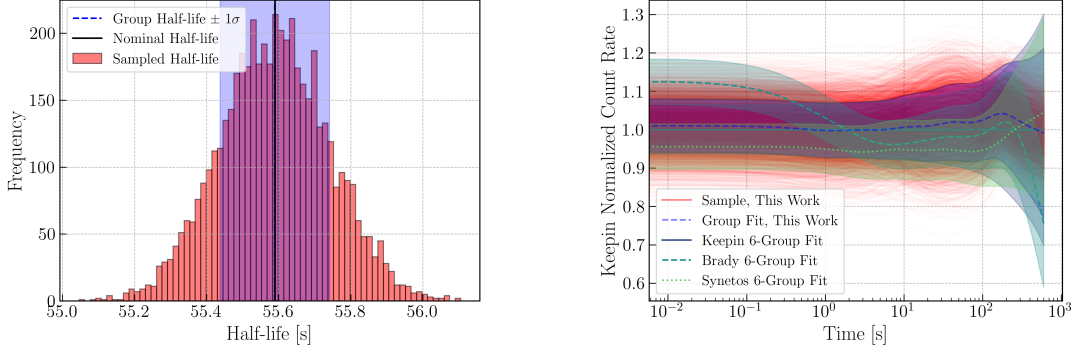
Fig. 1. The fraction of the delayed neutron yield from particular DNPs from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities. The five DNPs with the largest delayed neutron yield contributions supply over 50% of the total delayed neutrons per fission event. The remaining 215 DNPs provide approximately 48% of the total delayed neutron yield.

III.A. IAEA Beta-Delayed Neutron Emission Database

Figure 1 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). This equation shows that the largest yield contributors have large emission probabilities P_n and cumulative fission yields CFY . This doesn't necessarily indicate there will be a large concentration of the DNP, as it may be short lived, but rather for any given fission, that DNP has a higher probability to form and emit a delayed neutron. The total delayed neutron yield for this dataset is 0.0160 delayed neutrons per fission event. There are a total of 220 DNPs in this dataset, and five of them capture over 50% of the total delayed neutron yield.

Figure 2 shows the histogram for the sampled group one half-lives τ_1 as well as how the delayed neutron count rate compares to those using DNP group parameters from the literature [23, 6, 24]. The histogram shows that the group parameters follow a normal distribution as the nuclear data for each DNP is randomly sampled, with the group half-life as the mean of that normal distribution. Although not shown, the other group parameters follow this same trend. The Keepin normalized delayed neutron count rate shows how the delayed neutron count rate calculated in this work, both for every count rate from the sampled data and from the group fit using the calculated DNP group parameters compare. A value of 1 at any given time indicates the delayed neutron

count rate at that time perfectly matches the delayed neutron count rate from the Keepin et al. **DNP** group parameters. These results show that the Keepin et al. **DNP** group parameters agree well with group parameters calculated using the **IAEA** beta-delayed neutron emission database and JENDL-5 cumulative fission yields, with differences smaller than those from Brady and England and from Synetos and Williams.



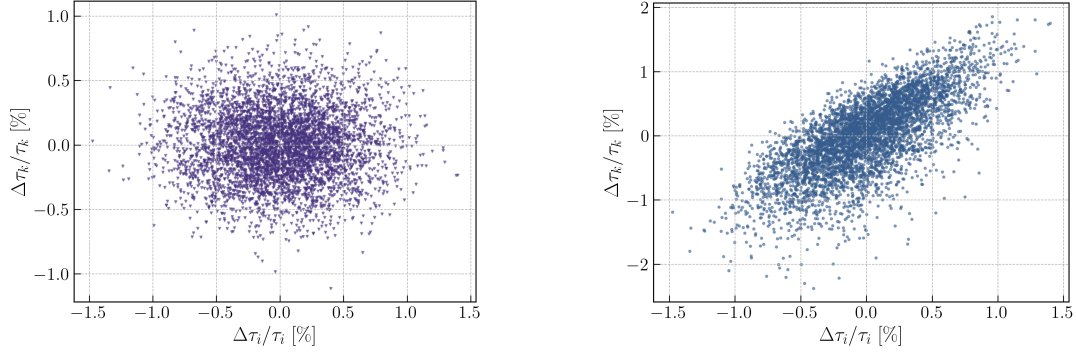
(a) The **DNP** group one half-life τ_1 evaluated based on the mean of the sampled data.

(b) The delayed neutron count rate as a function of time relative to the Keepin et al. **DNP** group parameter delayed neutron count rate for **DNP** six group parameters from this work, Brady and England, and Synetos and Williams [23, 6, 24].

Fig. 2. These plots show how the 5000 samples lead to a distribution of **DNP** group parameters and delayed neutron count rates from JENDL-5 cumulative fission yields and **IAEA** beta-delayed neutron emission database half-lives and emission probabilities. The **DNP** group one half-life τ_1 shows a normal distribution about a mean, while the Keepin et al. normalized count rate shows this work offers good agreement with the literature.

Figure 3 shows an example for how the **PCCs** are calculated using the 5000 samples. Each sample is represented as a single point at which the nuclear data is sampled from a normal distribution. Specifically for this figure, the samples are plotted as a function of the ^{137}I half-life. Figure 3(a) shows how the group one **DNP** half-life τ_1 changes as a function of the ^{137}I half-life. There is no clear increase or decrease in the group one half-life as the ^{137}I half-life varies, which is why the **PCC** is 0.005 for that combination. However, Figure 3(b) shows how the group two **DNP** half-life τ_2 has a much stronger correlation, resulting in a **PCC** of 0.73. These **PCC** values are calculated for the emission probability, concentration, and half-life for each group for each **DNP**.

Table III shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I **DNPs** and the six **DNP** groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general



(a) The relative change in the group one half-life τ_1 due to relative change in the ^{137}I half-life with a **PCC** of 0.005.

(b) The relative change in the group two half-life τ_2 due to relative change in the ^{137}I half-life with a **PCC** of 0.726.

Fig. 3. These plots show the sensitivity of the **DNP** group one and two half-lives to the ^{137}I half-life. The group one half-life τ_1 does not correlate with the ^{137}I half-life, resulting in a sample distribution with no discernible slope. The group two half-life τ_2 has a strong, positive linear correlation with the ^{137}I half-life with a **PCC** of 0.73. This indicates that changes to the ^{137}I half-life will have greater effects on the group two half-life than the group one half-life.

TABLE III

IAEA beta-delayed neutron emission database and JENDL-5 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0160 ± 0.0009	$[-]$
$\nu_d(K)$	0.0160 ± 0.0008	$[-]$
$\Delta\nu_d$	0.03 ± 122.82	$[pcm]$
$\bar{\tau}(I)$	9.1 ± 0.5	$[s]$
$\bar{\tau}(K)$	9.0 ± 0.4	$[s]$
$\Delta\bar{\tau}$	0.0 ± 0.6	$[s]$

behavior of the delayed neutrons.

Table IV shows the ten largest PCC_i terms, which reveals which **DNPs** have the greatest impact on the **DNP** group parameters $\nu_{d,k}$ and τ_k . This table shows that bromine alone has five isotopes that strongly affect the **DNP** group parameters, with ^{87}Br being the dominant isotope due to its long-half life that makes it the primary **DNP** for the group one parameters. Generally, **DNPs** will have a large PCC_i for one of two reasons: they affect several parameters moderately, such as ^{96}Rb which has an impact on group six as well as a smaller impact on groups five, four, three, and two as well; or they have a half-life that causes them to have a large impact on one specific group, such as ^{87}Br , which primarily affects group one.

Table V shows the ten largest U_i terms, which corresponds to which **DNPs** have the largest

TABLE IV

IAEA beta-delayed neutron emission database and JENDL-5 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{87}Br	3.678703
^{88}Br	3.397623
^{137}I	3.325644
^{96}Rb	3.277932
^{97}Rb	2.547497
^{94}Rb	2.157988
^{138}I	2.126428
^{89}Br	2.111075
^{90}Br	1.768819
^{91}Br	1.670366

TABLE V

IAEA beta-delayed neutron emission database and JENDL-5 nuclear data ten largest uncertainty-weighted PCCs.

Nuclide	U_i
^{105}Zr	0.281379
^{72}Co	0.268531
^{116}Ru	0.260299
^{105}Y	0.246537
^{75}Co	0.244795
^{68}Co	0.235227
^{116}Rh	0.231495
^{71}Co	0.229786
^{99}Kr	0.226118
^{70}Co	0.222885

uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k . These ten DNPs do not overlap at all with Table IV, indicating that the JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database have small relative uncertainties for the DNPs in Table IV as otherwise those DNPs would dominate Table V. Interestingly, it appears that cobalt has several isotopes that have large U_i values, indicating that element has large relative uncertainties in its nuclear data relative to its impact on the DNP group parameters.

Figure 4 gives a more detailed view on which DNP values specifically lead to the large U_i values. This offers slightly more insight than Table V, which sums over the various DNP values. This figure shows that the half-lives τ and emission probabilities P_n are the primary drivers of U_i , at least for the ten most impactful DNPs. This indicates that the JENDL-5 cumulative fission yields offer sufficiently relative uncertainties compared to the relative uncertainties from the IAEA beta-delayed neutron emission database.

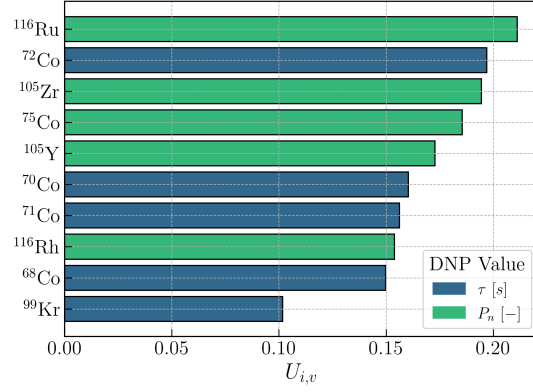
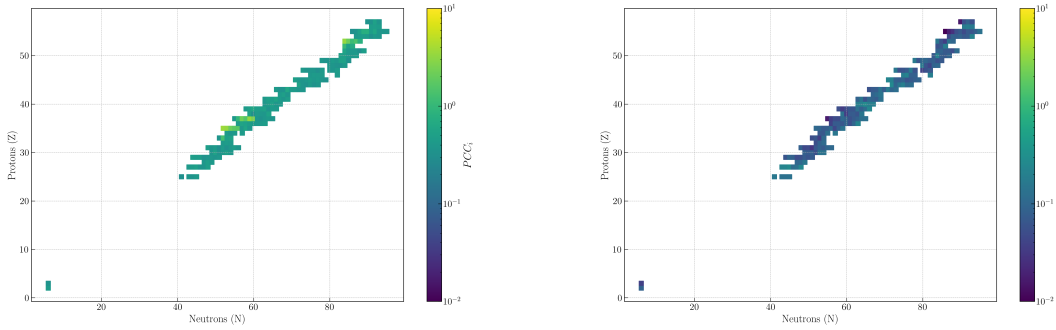


Fig. 4. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database half-lives and emission probabilities.

Figure 5 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i , because all of the relative uncertainties are less than 1. There do not appear to be any clear trends, though in Figure 5(a), some horizontal bands of larger PCC_i values can be seen for 35 protons, 37 protons, and 53 protons; these correspond to bromine, rubidium, and iodine, respectively.



(a) The PCC_i , or total PCC, calculated for each DNP.

(b) The U_i , or total uncertainty-weighted PCC, for each DNP.

Fig. 5. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using JENDL-5 cumulative fission yields and IAEA beta-delayed neutron emission database emission probabilities and half-lives.

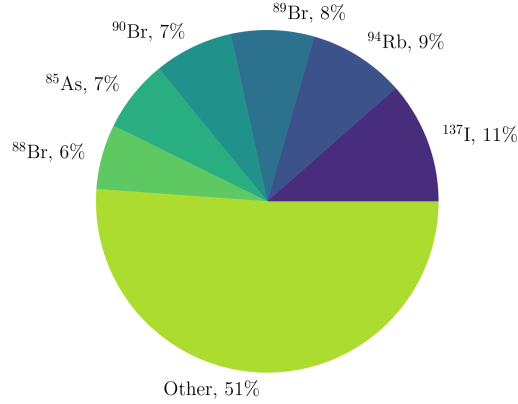


Fig. 6. The fraction of the delayed neutron yield from particular DNPs from ENDF/B-VII.1 nuclear data. The six DNPs with the largest delayed neutron yield contributions approximately 50% of the total delayed neutrons per fission event. The remaining 212 DNPs provide approximately 50% of the total delayed neutron yield.

III.B. ENDF/B-VII.1

Figure 6 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). These yields can be compared to those in Figure 1, which shows that in ENDF/B-VII.1, the yields of the top DNPs are relatively smaller. There is also the presence of ^{85}As , which did not appear in the top six DNPs when using the IAEA beta-delayed neutron emission database and JENDL-5 nuclear data.

Table VI shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general behavior of the delayed neutrons. However, the average half-life differs by approximately two seconds from Section III.A, while the total delayed neutron yield differs by approximately 300 pcm.

Table VII shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k . These results show that several DNPs have a stronger correlation than ^{87}Br , which is surprising considering its long half-life causes group one DNP group parameters to be more sensitive to it. However, differences in the emission probabilities, half-lives, and cumulative fission yields can cause DNPs such as ^{137}I to contribute more to DNP

TABLE VI
ENDF/B-VII.1 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0191 ± 0.0011	$[-]$
$\nu_d(K)$	0.0191 ± 0.0009	$[-]$
$\Delta\nu_d$	4.36 ± 135.24	$[pcm]$
$\bar{\tau}(I)$	7.2 ± 0.3	$[s]$
$\bar{\tau}(K)$	7.2 ± 0.3	$[s]$
$\Delta\bar{\tau}$	0.01 ± 0.4	$[s]$

TABLE VII
ENDF/B-VII.1 nuclear data ten largest [PCCs](#).

Nuclide	PCC_i
^{137}I	4.643721
^{137}Sb	4.612421
^{100}Rb	4.323455
^{87}Br	3.604110
^{85}As	3.025537
^{88}Br	2.177990
^{131}Cd	1.889176
^{88}As	1.880888
^{102}Y	1.311140
^{89}Br	1.242374

group two. For instance, Section [III.A](#) shows that the [PCC](#) for the group two half-life τ_2 from changes in the ^{137}I half-life is 0.726; but that same value calculated using the ENDF/B-VII.1 nuclear data is 0.922.

Table [VIII](#) shows the ten largest U_i terms, which corresponds to which [DNPs](#) have the largest uncertainties relative to their impact on the [DNP](#) group parameters $\nu_{d,k}$ and τ_k . There are several [DNPs](#) that exist in both Table [VII](#) and [VIII](#). These [DNPs](#) have a strong effect on the [DNP](#) group parameters and have large relative uncertainties. This means that, for ENDF/B-VII.1, these [DNPs](#) will lead to greater uncertainty in the calculated [DNP](#) group parameters and calculated delayed neutron emission rates.

Figure [7](#) gives a more detailed view on which [DNP](#) values specifically lead to the large U_i values. This offers slightly more insight than Table [VIII](#), which sums over the various [DNP](#) values. These results indicate that the half-lives are the primary sources of uncertainty that most affect the [DNP](#) group parameters. The top two [DNP](#) in Table [VIII](#) appear twice in this figure, as they have contributions from both the half-life nuclear data and their concentration. The nuclear data for ^{100}Rb and ^{137}Sb are the most problematic in ENDF/B-VII.1 when calculating [DNP](#) group

TABLE VIII

ENDF/B-VII.1 nuclear data ten largest uncertainty-weighted **PCCs**. Bold **DNPs** are present in Table VII as well, indicating they have a strong effect on **DNP** group parameters.

Nuclide	U_i
¹⁰⁰ Rb	1.365283
¹³⁷ Sb	1.181643
¹⁰⁰ Kr	0.429832
⁹⁹ Kr	0.370420
¹⁰² Y	0.320689
⁸⁷ As	0.307750
¹²⁴ Ag	0.275877
¹²⁴ Pd	0.254152
⁸⁴ As	0.249879
¹⁴⁷ Xe	0.241646

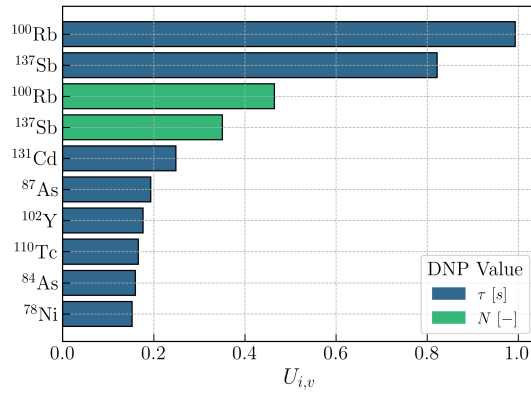
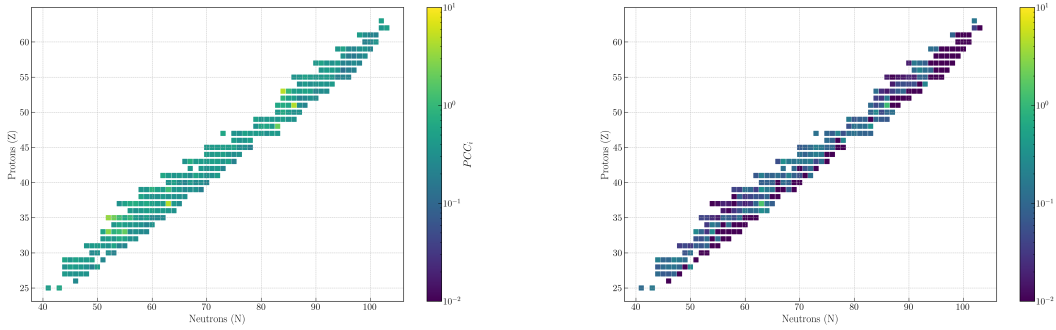


Fig. 7. The ten largest uncertainty-weighted **PCCs** $U_{i,v}$ sorted by **DNP** and associated **DNP** value from ENDF/B-VII.1 nuclear data.

parameters.

Figure 8 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i , because all of the relative uncertainties are less than 1. There are two points of interest that have large PCC_i and U_i values, ¹⁰⁰Rb and ¹³⁷Sb, which can be seen on both plots.



(a) The PCC_i , or total PCC , calculated for each DNP .

(b) The U_i , or total uncertainty-weighted PCC , for each DNP .

Fig. 8. Chart of nuclide plots showing the PCC s and uncertainty-weighted PCC s using ENDF/B-VII.1 nuclear data.

III.C. ENDF/B-VIII.0

Figure 9 shows the percentage of the total delayed neutron yield that comes from the six DNP s with the largest yield contribution, where the yield from each DNP is defined by Equation (1). There are 277 DNP s in ENDF/B-VIII.0, 59 more than in ENDF/B-VII.1. Although Figures 9 and 6 are similar, the main difference is the appearance of ^{86}As in the top six DNP yields.

Table IX shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNP s and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general behavior of the delayed neutrons. However, the average half-life differs by approximately 0.3 seconds from Section III.B, while the total delayed neutron yield differs by approximately 100 *pcm*. These differences put both results within uncertainty bounds of each other, though the ENDF/B-VIII.0 total delayed neutron yield and average half-life have even larger differences from the values calculated using the IAEA beta-delayed neutron emission database and JENDL-5 than the ENDF/B-VII.1 data.

Table X shows the ten largest PCC_i terms, which reveals which DNP s have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k . These results are similar to those presented in Section III.B using ENDF/B-VII.1 data. The main changes are due to the presence of arsenic-86 and 87, which was not present in the ten largest PCC s for ENDF/B-VII.1 data.

Table XI shows the ten largest U_i terms, which corresponds to which DNP s have the largest

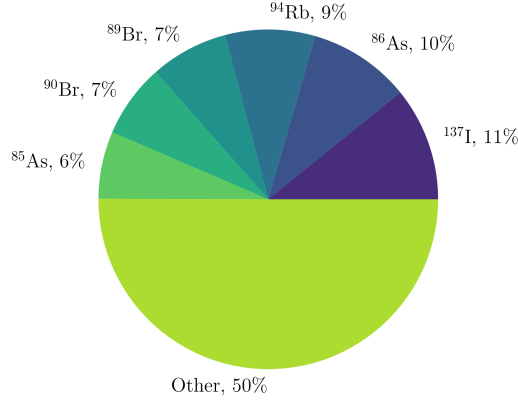


Fig. 9. The fraction of the delayed neutron yield from particular **DNPs** from ENDF/B-VIII.0 nuclear data. The six **DNPs** with the largest delayed neutron yield contributions approximately 50% of the total delayed neutrons per fission event. The remaining 271 **DNPs** provide approximately 50% of the total delayed neutron yield.

TABLE IX
ENDF/B-VIII.0 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0202 ± 0.0011	$[-]$
$\nu_d(K)$	0.0203 ± 0.0009	$[-]$
$\Delta\nu_d$	4.64 ± 142.02	$[pcm]$
$\bar{\tau}(I)$	6.9 ± 0.3	$[s]$
$\bar{\tau}(K)$	6.9 ± 0.3	$[s]$
$\Delta\bar{\tau}$	0.01 ± 0.4	$[s]$

TABLE X
ENDF/B-VIII.0 nuclear data ten largest **PCCs**.

Nuclide	PCC_i
^{137}Sb	5.052885
^{100}Rb	4.837795
^{137}I	4.468315
^{87}Br	3.350813
^{85}As	2.685888
^{88}As	2.107151
^{88}Br	2.093959
^{131}Cd	1.988529
^{86}As	1.862035
^{87}As	1.472684

TABLE XI

ENDF/B-VIII.0 nuclear data ten largest uncertainty-weighted **PCCs**. Bold **DNPs** are present in Table X as well, indicating they have a strong effect on **DNP** group parameters.

Nuclide	U_i
¹⁰⁰ Rb	1.510501
¹³⁷ Sb	1.274605
⁸⁷ As	0.380737
⁹⁹ Kr	0.355989
¹⁰² Y	0.291115
⁸⁴ As	0.268318
¹³¹ Cd	0.253599
⁹⁸ Y	0.245287
¹⁰⁴ Nb	0.240678
¹²⁴ Ag	0.240391

uncertainties relative to their impact on the **DNP** group parameters $\nu_{d,k}$ and τ_k . The results are very similar to those shown in Table VIII, though ⁸⁷As has taken the place of ¹⁰²Y for having a large **PCC** and relative uncertainty, appearing in both Table X and XI. Another new addition is ¹³¹Cd, which does not have any differences in cumulative fission yield, emission probability, or half-life between ENDF/B-VII.1 and ENDF/B-VIII.0. Instead the **PCCs** increased because of differences in other **DNPs**, as can be seen by comparing the ¹³¹Cd ENDF/B-VIII.0 PCC_i of 1.989 to the ENDF/B-VII.1 value of 1.889.

Figure 10 gives a more detailed view on which **DNP** values specifically lead to the large U_i values. This offers slightly more insight than Table XI, which sums over the various **DNP** values. The top four $U_{i,v}$ values are identical as those in Figure 7, indicating nuclear data for ¹⁰⁰Rb and ¹³⁷Sb remain problematic in ENDF/B-VIII.0 when calculating **DNP** group parameters.

Figure 11 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i , because all of the relative uncertainties are less than 1. The same points of interest as in Figure 8 are still relevant: ¹⁰⁰Rb and ¹³⁷Sb.

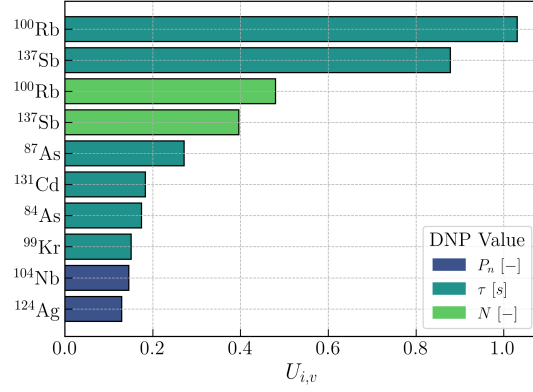
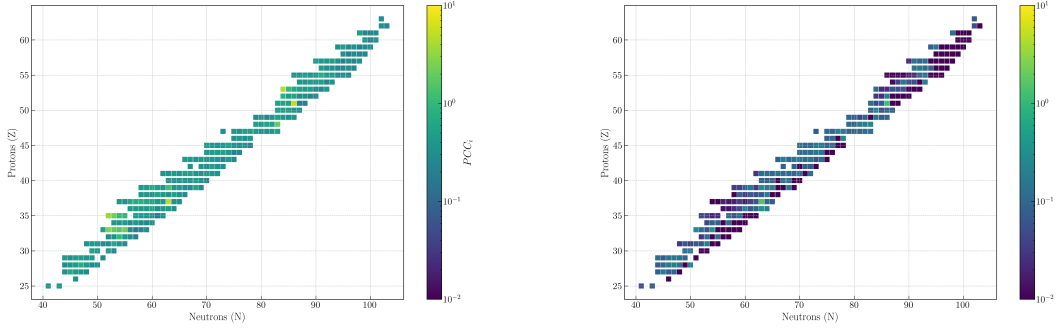


Fig. 10. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from ENDF/B-VII.1 nuclear data.



(a) The PCC_i , or total PCC, calculated for each DNP.

(b) The U_i , or total uncertainty-weighted PCC, for each DNP.

Fig. 11. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using ENDF/B-VIII.0 nuclear data.

III.D. JEFF-3.1.1

Figure 12 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). There are only 136 DNPs present in JEFF-3.1.1, and they have a total delayed neutron yield of 0.01477. This is much smaller than the total delayed neutron yields from other datasets, and thus leads each DNP to have a larger relative contribution to the delayed neutron yield.

Table XII shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The differences between both are well within the uncertainty bounds, indicating the six groups are sufficiently capturing the general

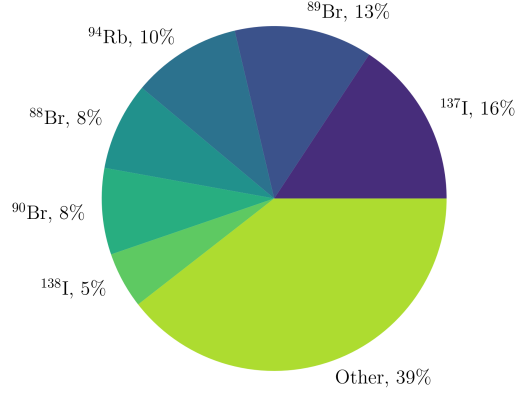


Fig. 12. The fraction of the delayed neutron yield from particular DNPs from JEFF-3.1.1 nuclear data. The five DNPs with the largest delayed neutron yield contributions approximately 56% of the total delayed neutrons per fission event. The remaining 131 DNPs provide approximately 44% of the total delayed neutron yield.

TABLE XII
JEFF-3.1.1 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0148 ± 0.0008	$[-]$
$\nu_d(K)$	0.0148 ± 0.0007	$[-]$
$\Delta\nu_d$	1.25 ± 108.88	$[pcm]$
$\bar{\tau}(I)$	9.2 ± 0.4	$[s]$
$\bar{\tau}(K)$	9.2 ± 0.4	$[s]$
$\Delta\bar{\tau}$	0.0 ± 0.6	$[s]$

behavior of the delayed neutrons. Although the total delayed neutron yield is smaller than other datasets, the average half-life is longer, indicating that the JEFF-3.1.1 data may be missing or under-predicting concentrations of short-lived DNP in particular.

Table XIII shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k . The top two DNPs, ^{137}I and ^{87}Br , strongly affect groups two and one as previously discussed, which is why they have the largest PCCs. Section III.A has a more detailed discussion on what effects increase PCCs.

Table XIV shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k . There are several DNPs that appear in Table XIV that also appear in Table XIII, indicated by the bold values. This highlighting indicates DNPs that strongly affect the DNP group parameters, meaning that, although their relative uncertainties may not be as large as the other DNPs in the table, the

TABLE XIII
JEFF-3.1.1 nuclear data ten largest **PCCs**.

Nuclide	PCC_i
^{137}I	3.619275
^{87}Br	3.498093
^{88}Br	3.323438
^{138}I	2.993978
^{96}Rb	2.727061
^{89}Br	2.304603
^{98}Y	2.055720
^{94}Rb	1.959133
^{91}Br	1.919183
^{90}Br	1.886707

TABLE XIV
JEFF-3.1.1 nuclear data ten largest uncertainty-weighted **PCCs**. Bold **DNPs** are present in Table **XIII** as well, indicating they have a strong effect on **DNP** group parameters.

Nuclide	U_i
^{98}Y	0.387086
^{93}Br	0.237471
^{147}Xe	0.214894
^{96}Rb	0.176298
^{138}Te	0.171012
^{94}Br	0.152192
^{91}Br	0.148399
^{158}Nd	0.144933
^{137}I	0.141580
^{111}Mo	0.134707

uncertainties they do have lead to large differences in the **DNP** group parameters and calculated delayed neutron emission rates.

Figure 13 gives a more detailed view on which **DNP** values specifically lead to the large U_i values. This offers slightly more insight than Table XIV, which sums over the various **DNP** values. For instance, these results show that ^{98}Y has a large contribution of its uncertainty-weighted sensitivity come from emission probability uncertainty, with a smaller contribution from its half-life uncertainty. It can also be seen that ^{137}I is not in Figure 13 even though it is in Table XIV, indicating that its uncertainty-weighted sensitivity comes from the sum of several smaller components rather than one large term. Specifically, 54.5% comes from emission probability uncertainty, 42.8% comes from concentration uncertainty, and the remainder comes from half-life uncertainty.

Figure 14 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not

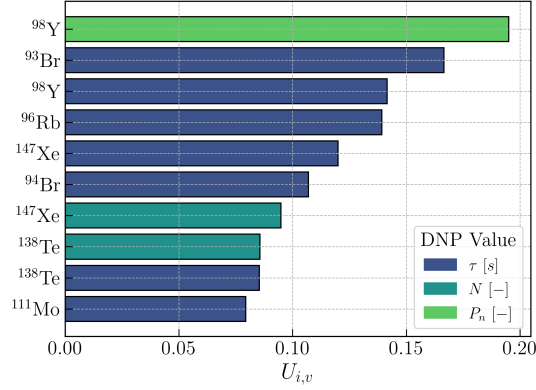
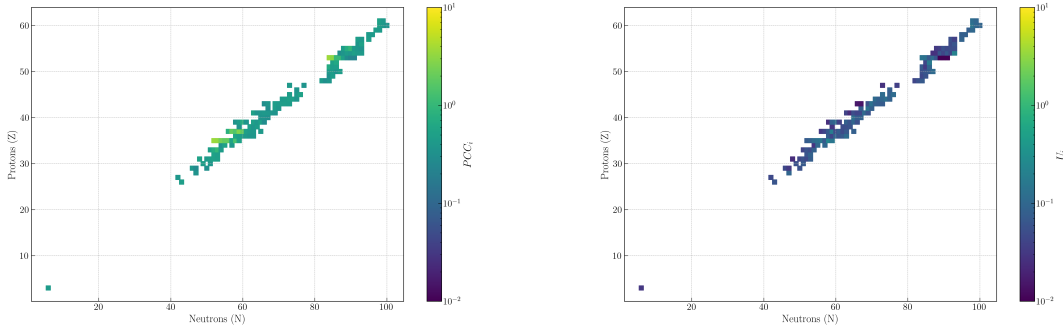


Fig. 13. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from ENDF/B-VII.1 nuclear data.

limit to only the top ten largest values, but instead show how all of the values compare among each other. The values of U_i are all smaller than those of PCC_i , because all of the relative uncertainties are less than 1. There are similar patterns as discussed in Section III.A, with horizontal peaks in the PCC_i for bromine, rubidium, and iodine.



(a) The PCC_i , or total PCC, calculated for each DNP.

(b) The U_i , or total uncertainty-weighted PCC, for each DNP.

Fig. 14. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using JEFF-3.1.1 nuclear data.

III.E. JENDL-5

Figure 15 shows the percentage of the total delayed neutron yield that comes from the six DNPs with the largest yield contribution, where the yield from each DNP is defined by Equation (1). There are 257 DNPs present in JENDL-5, and they have a total delayed neutron yield of

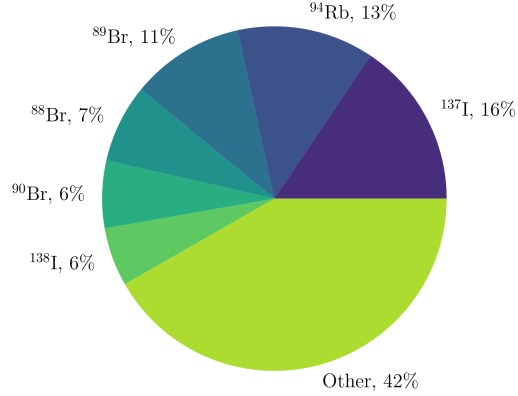


Fig. 15. The fraction of the delayed neutron yield from particular DNPs from JENDL-5 nuclear data. The five DNPs with the largest delayed neutron yield contributions approximately 52% of the total delayed neutrons per fission event.

TABLE XV
JENDL-5 nuclear data delayed neutron yield and average half-life values.

Parameter	Value	Units
$\nu_d(I)$	0.0160 ± 0.0009	$[-]$
$\nu_d(K)$	0.0160 ± 0.0008	$[-]$
$\Delta\nu_d$	0.09 ± 122.31	$[pcm]$
$\bar{\tau}(I)$	9.1 ± 0.5	$[s]$
$\bar{\tau}(K)$	9.1 ± 0.5	$[s]$
$\Delta\bar{\tau}$	0.0 ± 0.6	$[s]$

0.0160. This is consistent with the results in Section III.A, as the only difference between these datasets is the emission probability and half-life data.

Table XV shows how the total delayed neutron yield ν_d and average half-life $\bar{\tau}$ compares between the value from the I DNPs and the six DNP groups. The results agree well with the data in Section III.A.

Table XVI shows the ten largest PCC_i terms, which reveals which DNPs have the greatest impact on the DNP group parameters $\nu_{d,k}$ and τ_k . The same DNPs are in Table IV, though the exact values and order is slightly different due to half-life and emission probability differences.

Table XVII shows the ten largest U_i terms, which corresponds to which DNPs have the largest uncertainties relative to their impact on the DNP group parameters $\nu_{d,k}$ and τ_k . Four out of ten of these DNPs are also in Table V, though there are differences due to the emission probability and half-life differences. One notable difference is ^{102}Sr , which has a U_i value of 3.63×10^{13} . This value is significantly larger than any other value from any other dataset because it has an emission

TABLE XVI
JENDL-5 nuclear data ten largest PCCs.

Nuclide	PCC_i
^{87}Br	3.788538
^{88}Br	3.405082
^{137}I	3.373592
^{96}Rb	3.230474
^{97}Rb	2.406196
^{138}I	2.209843
^{94}Rb	2.202741
^{89}Br	2.123136
^{91}Br	1.779122
^{90}Br	1.733836

TABLE XVII
JENDL-5 nuclear data ten largest uncertainty-weighted PCCs.

Nuclide	U_i
^{102}Sr	-
^{142}Sb	0.283440
^{99}Kr	0.279344
^{105}Zr	0.270256
^{116}Rh	0.269286
^{103}Zr	0.267184
^{124}Ag	0.228787
^{68}Co	0.224881
^{134}Sn	0.207609
^{66}Mn	0.199226

probability of $(6.245 \times 10^{-17}) \pm 0.02$. This means that the relative uncertainty in Equation (17) works out to be 3.2×10^{-14} , which is exceedingly large. To continue conducting the analysis, the contribution of ^{102}Sr is ignored, treating the emission probability as 0 for this analysis.

Figure 16 gives a more detailed view on which DNP values specifically lead to the large U_i values. This offers slightly more insight than Table XVII, which sums over the various DNP values. These uncertainty-weighted sensitivities are closer to the ideal configuration: values are close to each other, and the uncertainties in half-lives, emission probability, and concentration each contribute some amount. There are no clear outliers in the data, though, similar to the results in Section III.A, the largest uncertainty-weighted sensitivity contributions come from emission probabilities, indicating a potential area of improvement.

Figure 17 shows the PCC_i and U_i plotted on the chart of the nuclides. These plots do not limit to only the top ten largest values, but instead show how all of the values compare among each

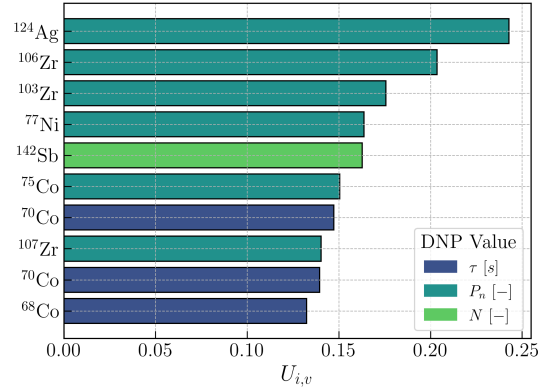
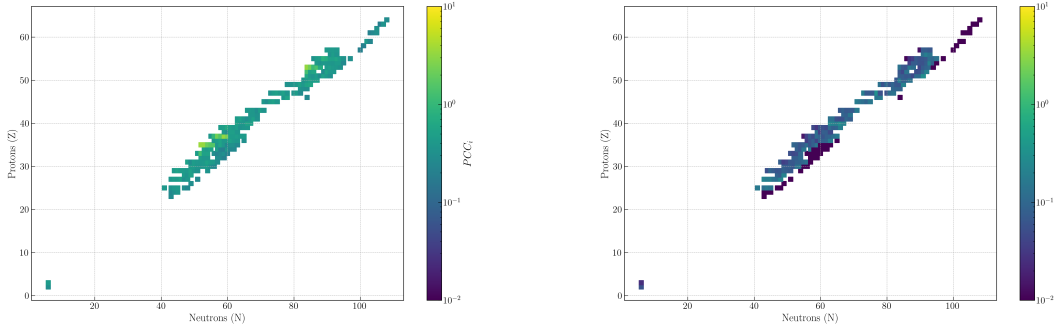


Fig. 16. The ten largest uncertainty-weighted PCCs $U_{i,v}$ sorted by DNP and associated DNP value from ENDF/B-VII.1 nuclear data.

other. The values of U_i are all smaller than those of PCC_i , because all of the relative uncertainties are less than 1. There are similar patterns as discussed in Section III.A, with horizontal peaks in the PCC_i for bromine, rubidium, and iodine.



(a) The PCC_i , or total PCC, calculated for each DNP.

(b) The U_i , or total uncertainty-weighted PCC, for each DNP.

Fig. 17. Chart of nuclide plots showing the PCCs and uncertainty-weighted PCCs using JENDL-5 nuclear data.

III.F. Hybrid

IV. CONCLUSIONS

- Summarize the main takeaways from the results (maybe a table indicating DNPs of interest for different datasets?)

Have a large table with all group parameters, yields, half-lives, etc. (maybe put full table in an appendix)

- Describe how improving these data will enable development of improved models of DNPs and kinetics models

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REFERENCES

- [1] M. C. BRADY, “Evaluation and application of delayed neutron precursor data,” PhD Thesis, Texas A&M University (1988).
- [2] B. PFEIFFER, K.-L. KRATZ, and P. MÖLLER, “Status of delayed-neutron precursor data: half-lives and neutron emission probabilities,” *Progress in nuclear energy*, **41**, 1-4, 39 (2002).
- [3] A. D’ANGELO and J. L. ROWLANDS, “A review of delayed neutron data for calculating effective delayed neutron fractions,” *Proc. PHYSOR*, 7–10 (2002).
- [4] G. RUDSTAM, P. FINCK, A. FILIP, A. D’ANGELO, and R. MCKNIGHT, “Delayed neutron data for the major actinides,” *A report by the Working Party on International Evaluation Co-operation of the OECD NEA Nuclear Science Committee*, **6** (2002).
- [5] W. WILSON and T. ENGLAND, “Delayed neutron study using ENDF/B-VI basic nuclear data,” *Progress in Nuclear Energy*, **41**, 1-4, 71 (2002).
- [6] M. BRADY and T. ENGLAND, “Delayed neutron data and group parameters for 43 fissioning systems,” *Nuclear Science and Engineering*, **103**, 2, 129 (1989).
- [7] R. MILLS, “Calculation of Delayed Neutron Emission from Fission and Fitting Results to Few-Group Approximations,” *Summary Report of the First Research Coordination Meeting on the Development of a Reference Database for Beta-Delayed Neutron Emission*, INDC(NDS)-0643, 65–68, International Atomic Energy Agency, IAEA, Vienna, Austria (2014)URL <https://www-nds.iaea.org/publications/indc/indc-nds-0643.pdf>, national Nuclear Laboratory, United Kingdom.
- [8] T. HUYNH and C. JOUANNE, “Calculation of delayed neutron yields for various libraries,” *Nuclear Data Sheets*, **118**, 476 (2014).
- [9] D. FOLIGNO, “New evaluation of delayed-neutron data and associated covariances,” Theses, CEA Cadarache, 13115 SAINT-PAUL-LEZ-DURANCE ; Aix Marseille Université, CNRS, Centrale Marseille, ED 353 Sciences pour l’ingénieur, Mécanique, physique, micro et nanoélectronique (2019)URL <https://theses.hal.science/tel-02469257>.

- [10] D. FOLIGNO, P. LECONTE, O. SEROT, O. LITAIZE, and A. CHEBBOUBI, “Summation calculation of delayed neutron yields for ^{235}U , ^{238}U and ^{239}Pu , based on various fission yield and neutron emission probability databases,” *EPJ Web of Conferences*, vol. 193, 03004 (2018); 10.1051/epjconf/201819303004., URL https://www.epj-conferences.org/articles/epjconf/abs/2018/28/epjconf_fission2017_03004/epjconf_fission2017_03004.html, publisher: EDP Sciences.
- [11] P. DIMITRIOU, I. DILLMANN, B. SINGH, V. PIKSAIKIN, K. RYKACZEWSKI, J. TAIN, A. ALGORA, K. BANERJEE, I. BORZOV, D. CANO-OTT, S. CHIBA, M. FALLOT, D. FOLIGNO, R. GRZYWACZ, X. HUANG, T. MARKETIN, F. MINATO, G. MUKHERJEE, B. RASCO, A. SONZOGNI, M. VERPELLI, A. EGOROV, M. ESTIENNE, L. GIOT, D. GREMYACHKIN, M. MADURGA, E. MCCUTCHAN, E. MENDOZA, K. MITROFANOV, M. NARBONNE, P. ROMOJARO, A. SANCHEZ-CABALLERO, and N. SCIELZO, “Development of a Reference Database for Beta-Delayed Neutron Emission,” *Nuclear Data Sheets*, **173**, 144 (2021); <https://doi.org/10.1016/j.nds.2021.04.006>., URL <https://www.sciencedirect.com/science/article/pii/S0090375221000168>, special Issue on Nuclear Reaction Data.
- [12] A. J. PLOMPEN, O. CABELLOS, C. DE SAINT JEAN, M. FLEMING, A. ALGORA, M. ANGELONE, P. ARCHIER, E. BAUGE, O. BERSILLON, A. BLOKHIN ET AL., “The joint evaluated fission and fusion nuclear data library, JEFF-3.3,” *The European Physical Journal A*, **56**, 1 (2020).
- [13] M. B. CHADWICK, M. HERMAN, P. OBLOŽINSKÝ, M. E. DUNN, Y. DANON, A. KAHLER, D. L. SMITH, B. PRITYCHENKO, G. ARBANAS, R. ARCILLA ET AL., “ENDF/B-VII. 1 nuclear data for science and technology: cross sections, covariances, fission product yields and decay data,” *Nuclear data sheets*, **112**, 12, 2887 (2011).
- [14] D. BROWN, M. CHADWICK, R. CAPOTE, A. KAHLER, A. TRKOV, M. HERMAN, A. SONZOGNI, Y. DANON, A. CARLSON, M. DUNN, D. SMITH, G. HALE, G. ARBANAS, R. ARCILLA, C. BATES, B. BECK, B. BECKER, F. BROWN, R. CASPERSON, J. CONLIN, D. CULLEN, M.-A. DESCALLE, R. FIRESTONE, T. GAINES, K. GUBER, A. HAWARI, J. HOLMES, T. JOHNSON, T. KAWANO, B. KIEDROWSKI, A. KONING, S. KOPECKY, L. LEAL, J. LESTONE, C. LUBITZ, J. M. DAMIÁN, C. MATTOON, E. MCCUTCHAN, S. MUGHABGHAB, P. NAVRATIL,

- D. NEUDECKER, G. NOBRE, G. NOGUERE, M. PARIS, M. PIGNI, A. PLOMPEN, B. PRITYCHENKO, V. PRONYAEV, D. ROUBTSOV, D. ROCHMAN, P. ROMANO, P. SCHILLEBEECKX, S. SIMAKOV, M. SIN, I. SIRAKOV, B. SLEAFORD, V. SOBES, E. SOUKHOVITSKII, I. STETCU, P. TALOU, I. THOMPSON, S. VAN DER MARCK, L. WELSER-SHERRILL, D. WIARDA, M. WHITE, J. WORMALD, R. WRIGHT, M. ZERKLE, G. ŽEROVNIK, and Y. ZHU, “ENDF/B-VIII.0: The 8th Major Release of the Nuclear Reaction Data Library with CIELO-project Cross Sections, New Standards and Thermal Scattering Data,” *Nuclear Data Sheets*, **148**, 1 (2018); <https://doi.org/10.1016/j.nds.2018.02.001>, URL <https://www.sciencedirect.com/science/article/pii/S0090375218300206>, special Issue on Nuclear Reaction Data.
- [15] O. IWAMOTO, N. IWAMOTO, S. KUNIEDA, F. MINATO, S. NAKAYAMA, Y. ABE, K. TSUBAKIHARA, S. OKUMURA, C. ISHIZUKA, T. YOSHIDA ET AL., “Japanese evaluated nuclear data library version 5: JENDL-5,” *Journal of Nuclear Science and Technology*, **60**, 1, 1 (2023).
- [16] ERIC O. LEBIGOT. A, “Uncertainties: a Python package for calculations with uncertainties,” (2024).
- [17] P. VIRTANEN, R. GOMMERS, T. E. OLIPHANT, M. HABERLAND, T. REDDY, D. COURNAPEAU, E. BUROVSKI, P. PETERSON, W. WECKESSER, J. BRIGHT, S. J. VAN DER WALT, M. BRETT, J. WILSON, K. J. MILLMAN, N. MAYOROV, A. R. J. NELSON, E. JONES, R. KERN, E. LARSON, C. J. CAREY, Í. POLAT, Y. FENG, E. W. MOORE, J. VANDERPLAS, D. LAXALDE, J. PERKTOLD, R. CIMRMAN, I. HENRIKSEN, E. A. QUINTERO, C. R. HARRIS, A. M. ARCHIBALD, A. H. RIBEIRO, F. PEDREGOSA, P. VAN MULBREGT, and SciPy 1.0 CONTRIBUTORS, “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python,” *Nature Methods*, **17**, 261 (2020); 10.1038/s41592-019-0686-2.
- [18] J. P. CHANDLER, “STEPIT—Finds Local Minima of a Smooth Function of Several Parameters,” *Behavioral Science*, **14**, 81 (1969)URL https://www.researchgate.net/publication/243773790_STEPIT-Finds_local_minima_of_a_smooth_function_of_several_parameters.
- [19] R. WALDO, R. KARAM, and R. MEYER, “Delayed neutron yields: Time dependent measurements and a predictive model,” *Physical Review C*, **23**, 3, 1113 (1981).

- [20] P. R. BEVINGTON and D. K. ROBINSON, *Data Reduction and Error Analysis for the Physical Sciences*, McGraw-Hill (2003).
- [21] N. W. TOURAN, J. GILLELAND, G. T. MALMGREN, C. WHITMER, and W. H. GATES, “Computational Tools for the Integrated Design of Advanced Nuclear Reactors,” *Engineering*, **3**, 4, 518 (2017); 10.1016/J.ENG.2017.04.016., URL <http://www.sciencedirect.com/science/article/pii/S2095809917306124>.
- [22] A. SANTAMARINA, D. BERNARD, P. BLAISE, M. COSTE, A. COURCELLE, T. HUYNH, C. JOUANNE, P. LECONTE, O. LITAIZE, S. MENGELLE ET AL., “The JEFF-3.1.1 nuclear data library,” *JEFF report*, **22**, 10.2, 2 (2009).
- [23] G. R. KEEPIN, T. WIMETT, and R. ZEIGLER, “Delayed neutrons from fissionable isotopes of uranium, plutonium and thorium,” *Journal of Nuclear Energy (1954)*, **6**, 1-2, IN2 (1957).
- [24] S. SYNETOS and J. WILLIAMS, “Delayed neutron yield and decay constants for thermal neutron-induced fission of ^{235}U ,” *Nucl. Energy (Inst. Nucl. Eng.)*, **22**, 4 (1983).